Purification, Thermal Ensembles, and `conserve="Sz"` in TenPy

Main conclusion

For the J1-J2 triangular Heisenberg model in `finite_T_J1J2.py`, using

`conserve="Sz"`

does *not* mean that the physical thermal ensemble is restricted to one fixed magnetization sector. In the purification calculation used in the code, the physical system is doubled by an auxiliary system. The U(1) charge constraint is imposed on the *doubled purified state*, where the auxiliary charges compensate the physical charges. Therefore all physical total-S^z sectors remain present in the thermal trace.

The line

`PurificationMPS.from_infiniteT(...)`

constructs the grand-canonical infinite-temperature purification. I checked the installed TenPy source used in this environment, TenPy version `1.1.0`: `PurificationMPS.from_infiniteT` is the grand-canonical constructor, whereas TenPy has a separate constructor for a fixed charge sector,

`PurificationMPS.from_infiniteT_canonical(..., charge_sector=...)`.

Thus, for the Hamiltonian currently implemented,

$$H = \sum_{\langle i,j\rangle} J_1\, \boldsymbol{S}_i \cdot \boldsymbol{S}_j + \sum_{\langle\langle i,j\rangle\rangle} J_2\, \boldsymbol{S}_i \cdot \boldsymbol{S}_j + F_z \sum_i S_i^z,$$

using `conserve="Sz"` is physically sensible and usually preferable. It is a symmetry optimization. It is not a projection to the physical $S^z_{\text{tot}} = 0$, $S^z_{\text{tot}} = -N/2$, or any other fixed magnetization sector.

1 Thermal density matrices from the beginning

Let the physical Hilbert space of N spin-1/2 sites be

$$\mathcal{H}_{\text{phys}} = \bigotimes_{i=1}^{N} \mathbb{C}^2.$$

The thermal density matrix at inverse temperature $\beta = 1/T$ is

$$\rho_\beta = \frac{e^{-\beta H}}{Z(\beta)}, \qquad Z(\beta) = \mathrm{Tr}_{\mathcal{H}_{\text{phys}}}\left(e^{-\beta H}\right).$$

For any physical observable O,

$$\langle O\rangle_\beta = \mathrm{Tr}_{\mathcal{H}_{\mathrm{phys}}}(\rho_\beta O) = \frac{\mathrm{Tr}_{\mathcal{H}_{\mathrm{phys}}}\left(e^{-\beta H}O\right)}{\mathrm{Tr}_{\mathcal{H}_{\mathrm{phys}}}\left(e^{-\beta H}\right)}.$$

If the Hamiltonian conserves total magnetization,

$$[H, S^z_{\mathrm{tot}}] = 0, \qquad S^z_{\mathrm{tot}} = \sum_i S^z_i,$$

then the physical Hilbert space decomposes into charge sectors:

$$\mathcal{H}_{\mathrm{phys}} = \bigoplus_M \mathcal{H}^{(M)}_{\mathrm{phys}},$$

where M is the total S^z. In this case the grand-canonical thermal partition function, with no fixed magnetization constraint, is

$$Z(\beta) = \sum_M \mathrm{Tr}_{\mathcal{H}^{(M)}_{\mathrm{phys}}}\left(e^{-\beta H_M}\right).$$

Here H_M is the Hamiltonian restricted to sector M.

A fixed-sector, or canonical-in-S^z, calculation would instead use

$$Z_M(\beta) = \mathrm{Tr}_{\mathcal{H}^{(M)}_{\mathrm{phys}}}\left(e^{-\beta H_M}\right)$$

for a single chosen M. This is *not* what `PurificationMPS.from_infiniteT` does.

If $F_z \neq 0$, then the Hamiltonian includes a longitudinal field

$$F_z S^z_{\mathrm{tot}}.$$

This biases the magnetization thermally, but it still does not fix S^z_{tot}. The trace still sums over all M, with field-dependent Boltzmann weights.

2 What purification does

Purification rewrites a mixed density matrix on $\mathcal{H}_{\mathrm{phys}}$ as a pure state on a larger Hilbert space

$$\mathcal{H}_{\mathrm{phys}} \otimes \mathcal{H}_{\mathrm{aux}} = \mathcal{H}_{\mathrm{phys}} \otimes \mathcal{H}_{\mathrm{aux}}.$$

The auxiliary space $\mathcal{H}_{\mathrm{aux}}$ is a copy of the physical Hilbert space.

Choose an orthonormal basis $\{|n\rangle\}$ of $\mathcal{H}_{\mathrm{phys}}$. The infinite temperature purification is the maximally entangled state

$$|\Psi_\infty\rangle = \frac{1}{\sqrt{\dim \mathcal{H}_{\mathrm{phys}}}} \sum_n |n\rangle_{\mathrm{phys}} \otimes |n\rangle_{\mathrm{aux}}.$$

Tracing out the auxiliary space gives

$$\mathrm{Tr}_{\mathrm{aux}}\left(|\Psi_\infty\rangle\langle\Psi_\infty|\right) = \frac{1}{\dim \mathcal{H}_{\mathrm{phys}}} \mathbb{1}_{\mathrm{phys}}.$$

This is the infinite-temperature density matrix.

At finite temperature, imaginary-time evolution is applied only to the physical system:

$$|\Psi_\beta\rangle = \left(e^{-\beta H/2} \otimes \mathbb{1}_{\mathrm{aux}}\right) |\Psi_\infty\rangle.$$

Then

$$\mathrm{Tr}_{\mathrm{aux}} \left(|\Psi_\beta\rangle\langle\Psi_\beta|\right) \propto e^{-\beta H}.$$

The normalization is

$$\langle\Psi_\beta|\Psi_\beta\rangle = \mathrm{Tr}_{\mathcal{H}_{\mathrm{phys}}} \left(e^{-\beta H}\right) = Z(\beta).$$

Therefore

$$\langle O\rangle_\beta = \frac{\langle\Psi_\beta|O|\Psi_\beta\rangle}{\langle\Psi_\beta|\Psi_\beta\rangle}.$$

3 One spin: the simplest example

For one spin-1/2, the physical basis is

$$|\uparrow\rangle, \qquad |\downarrow\rangle.$$

The infinite-temperature purification is

$$|\Psi_\infty\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\rangle_{\mathrm{phys}}|\uparrow\rangle_{\mathrm{aux}} + |\downarrow\rangle_{\mathrm{phys}}|\downarrow\rangle_{\mathrm{aux}}\right).$$

Tracing over the auxiliary spin gives

$$\rho_\infty = \mathrm{Tr}_{\mathrm{aux}} |\Psi_\infty\rangle\langle\Psi_\infty| = \frac{1}{2}\left(|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|\right) = \frac{1}{2}\mathbb{1}.$$

Both physical $S^z = +1/2$ and $S^z = -1/2$ states are present. No physical sector has been removed.

4 Many spins: all physical magnetization sectors are included

For N spins,

$$|\Psi_\infty\rangle = \frac{1}{\sqrt{2^N}} \sum_{\sigma_1,\ldots,\sigma_N} |\sigma_1,\ldots,\sigma_N\rangle_{\mathrm{phys}} \otimes |\sigma_1,\ldots,\sigma_N\rangle_{\mathrm{aux}}.$$

The sum runs over all 2^N spin configurations. These configurations include every physical total magnetization sector:

$$M = -\frac{N}{2}, -\frac{N}{2}+1, \ldots, \frac{N}{2}.$$

Thus the infinite-temperature state is not a fixed-M state of the physical system. It is the full identity density matrix after tracing out the auxiliary system.

5 What `conserve="Sz"` means in an ordinary MPS

In an ordinary pure-state MPS without purification, using a U(1) symmetry often does mean that the represented physical wavefunction has a fixed total charge. For spin systems, the charge is usually

$$Q = 2S^z_{\mathrm{tot}}.$$

For example, an ordinary MPS with fixed total charge $Q = 0$ represents a pure state in the physical $S^z_{\text{tot}} = 0$ sector.

This intuition is the source of the confusion. Purification is different because the MPS represents a vector in the doubled space

$$\mathcal{H}_{\text{phys}} \otimes \mathcal{H}_{\text{aux}},$$

not a wavefunction only in $\mathcal{H}_{\text{phys}}$.

6 What is the local MPS tensor?

An MPS is just a factorized way to store the coefficients of a many-body state. For an ordinary spin chain,

$$|\psi\rangle = \sum_{\sigma_1,\ldots,\sigma_N} C_{\sigma_1,\ldots,\sigma_N} |\sigma_1, \ldots, \sigma_N\rangle.$$

The coefficient tensor $C_{\sigma_1,\ldots,\sigma_N}$ is exponentially large. An MPS rewrites it as a product of local tensors:

$$C_{\sigma_1,\ldots,\sigma_N} = \sum_{\alpha_1,\ldots,\alpha_{N-1}} B_1^{\sigma_1}{}_{\alpha_0,\alpha_1} B_2^{\sigma_2}{}_{\alpha_1,\alpha_2} \cdots B_N^{\sigma_N}{}_{\alpha_{N-1},\alpha_N}.$$

For finite boundary conditions, α_0 and α_N have dimension one. The indices α_i are virtual or bond indices. The physical index σ_i labels the local spin state.

In TenPy, the local MPS tensors are usually called B_i. This B is not a magnetic field; it is the conventional name for the local MPS tensor. Depending on the canonical form, TenPy may internally store tensors in A, B, or mixed form, but the conceptual object is the same: a local tensor with one physical leg and two virtual legs.

For purification, each site has two local legs:

$$p_i = \text{physical spin index}, \qquad q_i = \text{auxiliary spin index}.$$

Thus a purified MPS stores coefficients

$$C_{p_1,q_1;\ldots;p_N,q_N}$$

through local tensors

$$B_i^{p_i,q_i}{}_{\alpha_{i-1},\alpha_i}.$$

Equivalently,

$$|\Psi\rangle = \sum_{\{p_i\},\{q_i\}} C_{p_1,q_1;\ldots;p_N,q_N} |p_1, \ldots, p_N\rangle_{\text{phys}} \otimes |q_1, \ldots, q_N\rangle_{\text{aux}}.$$

7 The infinite-temperature local tensor

At infinite temperature, the purified state should be

$$|\Psi_\infty\rangle = \frac{1}{\sqrt{2^N}} \sum_{\sigma_1,\ldots,\sigma_N} |\sigma_1, \ldots, \sigma_N\rangle_{\text{phys}} \otimes |\sigma_1, \ldots, \sigma_N\rangle_{\text{aux}}.$$

This means that the physical and auxiliary configurations are identical in each nonzero term:

$$p_i = q_i \quad \text{for every site } i.$$

Therefore the coefficient of a doubled configuration is

$$C_{p_1,q_1;\ldots;p_N,q_N} = \frac{1}{\sqrt{2^N}} \prod_{i=1}^{N} \delta_{p_i,q_i}.$$

This is why the local infinite-temperature tensor is diagonal in p and q:

$$B_i^{p_i,q_i} = \frac{1}{\sqrt{2}} \delta_{p_i,q_i}$$

up to the trivial left and right virtual legs present at $\beta = 0$. For one site, explicitly,

$$B^{\uparrow,\uparrow} = \frac{1}{\sqrt{2}}, \qquad B^{\downarrow,\downarrow} = \frac{1}{\sqrt{2}}, \qquad B^{\uparrow,\downarrow} = B^{\downarrow,\uparrow} = 0.$$

This is exactly the tensor version of

$$|\Psi_\infty\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\rangle_{\text{phys}} |\uparrow\rangle_{\text{aux}} + |\downarrow\rangle_{\text{phys}} |\downarrow\rangle_{\text{aux}} \right)$$

on each site.

In the installed TenPy source, version `1.1.0`, the relevant file is

```
tenpy/networks/purification_mps.py
```

The relevant code is:

```
for i in range(L):
    p_leg = sites[i].leg
    B = npc.diag(1.0, p_leg, dtype, ['p', 'q']) / sites[i].dim ** 0.5
    # leg `q` has the physical leg with opposite `qconj`
    B = B.add_trivial_leg(0, label='vL', qconj=+1).add_trivial_leg(1, label='vR', qconj=-1)
    Bs[i] = B
```

There are two important facts in this block.

(i) `npc.diag(..., ['p', 'q'])` makes a diagonal tensor between the physical leg p and the auxiliary leg q. This is the code version of

$$B_i^{p_i,q_i} \propto \delta_{p_i,q_i}.$$

(ii) The comment says that the auxiliary leg q has the physical leg with opposite `qconj`. This is TenPy's way of saying that the auxiliary leg is the dual, or bra-like, copy of the physical leg for charge bookkeeping.

For the spin-1/2 site itself, the relevant part of TenPy's `SpinHalfSite` source is:

```
if conserve == 'Sz':
    chinfo = npc.ChargeInfo([1], ['2*Sz'])
    leg = npc.LegCharge.from_qflat(chinfo, [1, -1])
```

Thus the local physical charges are $+1$ for $|\uparrow\rangle$ and -1 for $|\downarrow\rangle$, using the convention $Q = 2S^z$. The diagonal construction gives $\delta_{p,q}$. The opposite `qconj` on the auxiliary leg gives the charge cancellation explained next.

8 What `conserve="Sz"` means in purification

In TenPy's purification MPS, each site tensor has a physical leg and an auxiliary leg. Call their local basis labels p and q. For `SpinHalfSite` with `conserve="Sz"`, the physical charges are

$$Q(|\uparrow\rangle) = +1, \qquad Q(|\downarrow\rangle) = -1,$$

where $Q = 2S^z$.

The important implementation detail is that the auxiliary leg q is used with the opposite charge orientation. A nonzero tensor element obeys the charge selection rule

$$Q(\alpha_i) = Q(\alpha_{i-1}) + Q(p_i) - Q(q_i),$$

up to TenPy's internal sign convention for incoming and outgoing legs. The physical meaning is simple: the virtual bond charge tracks the accumulated difference

$$Q_{\text{bond}}(i) = \sum_{j \le i} \left[Q(p_j) - Q(q_j)\right].$$

For a finite MPS with trivial boundary charges, the total doubled charge is

$$\sum_{j=1}^{N} \left[Q(p_j) - Q(q_j)\right] = 0.$$

This is the charge that is conserved by the purification MPS with `conserve="Sz"`:

$$Q_{\text{doubled}} = Q_{\text{phys}} - Q_{\text{aux}}.$$

It is *not* the physical charge Q_{phys} alone.

At $\beta = 0$, the diagonal tensor $B_{p,q} \propto \delta_{p,q}$ makes every site locally neutral in this doubled sense:

$$Q(p_i) - Q(q_i) = 0.$$

Thus all virtual bond charges are zero at infinite temperature. This does not mean all physical spins are in the same S^z sector. It means that every physical configuration is paired with an auxiliary configuration of the same charge.

Because the auxiliary leg has the opposite effective charge, the physical charge and auxiliary charge cancel:

$$Q_{\text{phys}}(p) + Q_{\text{aux,effective}}(q) = 0 \quad \text{when } p = q.$$

For example,

$$|\uparrow\rangle_{\text{phys}} |\uparrow\rangle_{\text{aux}} \quad \text{has} \quad Q_{\text{phys}} = +1, \quad Q_{\text{aux,effective}} = -1,$$

so the doubled charge is zero. Similarly,

$$|\downarrow\rangle_{\text{phys}} |\downarrow\rangle_{\text{aux}} \quad \text{has} \quad Q_{\text{phys}} = -1, \quad Q_{\text{aux,effective}} = +1,$$

so the doubled charge is also zero.

Therefore the purified MPS can have total doubled charge zero while still containing all physical magnetization sectors.

Another way to say the same thing is:

$$\texttt{conserve="Sz"} \quad \Longrightarrow \quad \text{TenPy conserves charge on the doubled purified MPS.}$$

It does not imply

$$\text{physical } S^z_{\text{tot}} \text{ is fixed to one chosen value.}$$

Only if one explicitly uses a canonical fixed-charge purification is a single physical charge sector selected, for example

`from_infiniteT_canonical`.

9 Why the auxiliary leg has opposite charge orientation

The reason is that purification is closely related to vectorizing an operator. Write a physical operator X as

$$X = \sum_{p,q} X_{p,q} \, |p\rangle\langle q|.$$

The index p is a ket index. The index q is a bra index. Under a U(1) rotation generated by charge Q,

$$|p\rangle \mapsto e^{i\theta Q(p)}|p\rangle, \qquad \langle q| \mapsto e^{-i\theta Q(q)}\langle q|.$$

Therefore the operator basis element transforms as

$$|p\rangle\langle q| \mapsto e^{i\theta[Q(p)-Q(q)]}|p\rangle\langle q|.$$

Its charge is not $Q(p) + Q(q)$. Its charge is

$$Q(p) - Q(q).$$

Purification represents this operator-like object as a vector in a doubled Hilbert space:

$$|X\rangle\rangle = \sum_{p,q} X_{p,q}|p\rangle_{\text{phys}} \otimes |q\rangle_{\text{aux}}.$$

To preserve the operator charge $Q(p) - Q(q)$, the auxiliary leg must carry the opposite charge orientation. This is why the auxiliary leg is not treated as a second independent physical spin with the same charge sign. It represents the bra index of the density matrix.

The infinite-temperature operator is the identity,

$$X = \mathbb{1} = \sum_{p} |p\rangle\langle p|.$$

Vectorizing it gives

$$|\mathbb{1}\rangle\rangle = \sum_{p} |p\rangle_{\text{phys}} \otimes |p\rangle_{\text{aux}}.$$

Every term has doubled charge

$$Q(p) - Q(p) = 0.$$

This is exactly why the local tensor is diagonal:

$$B_{p,q} \propto \delta_{p,q}.$$

10 Why one still sets `conserve="Sz"` by hand

The line in TenPy's purification code

```
# leg ‘q‘ has the physical leg with opposite ‘qconj‘
```

does not by itself create a nontrivial S^z conservation law. It only says how to orient the charge of the auxiliary leg relative to the physical leg. There must already be a nontrivial charge on the physical leg.

That physical-leg charge is created by the site definition. In your model this happens here:

```
site = SpinHalfSite(conserve=conserve)
```

If one passes

```
conserve="Sz"
```

then TenPy constructs the physical leg with charges

$$Q(|\uparrow\rangle) = +1, \qquad Q(|\downarrow\rangle) = -1.$$

Then `from_infiniteT` uses the same physical leg for p, and the opposite orientation for q. The result is a nontrivial doubled U(1) symmetry:

$$Q_{\mathrm{doubled}} = Q_{\mathrm{phys}} - Q_{\mathrm{aux}}.$$

If instead one passes

```
conserve=None
```

then `SpinHalfSite` uses a trivial charge leg. In that case, the auxiliary leg still has the opposite `qconj`, but the charge itself is trivial. Schematically,

$$Q(|\uparrow\rangle) = 0, \qquad Q(|\downarrow\rangle) = 0.$$

Then

$$Q_{\mathrm{doubled}} = 0 - 0 = 0$$

for every tensor element, and the U(1) block structure is absent. The calculation can still represent the same physics, but it stores tensors densely instead of using the S^z block sparsity.

So there are two separate ingredients:

(i) `conserve="Sz"` on `SpinHalfSite`: define a nontrivial U(1) charge on the local physical basis.

(ii) opposite `qconj` on the purification auxiliary leg: make that charge act as a bra/dual charge, so the purified tensor tracks $Q_{\mathrm{phys}} - Q_{\mathrm{aux}}$.

Both are needed for a useful S^z-symmetric purification MPS. The first supplies the charge labels. The second orients the auxiliary copy correctly.

This is why `conserve="Sz"` is not redundant. Without it, TenPy still builds a grand-canonical infinite-temperature purification, but it does not use the nontrivial S^z symmetry. With it, TenPy builds the same physical grand-canonical purification in a symmetry-resolved tensor representation.

11 Why TenPy's purification example does not use `conserve="Sz"`

The official TenPy purification example uses the transverse-field Ising chain:

```
from tenpy.models.tf_ising import TFIChain

model_params = dict(L=L, J=1.0, g=1.2)
M = TFIChain(model_params)
psi = PurificationMPS.from_infiniteT(M.lat.mps_sites(), bc=bc)
```

This is a good minimal example of purification, but it is not an example of a Hamiltonian with U(1) S^z conservation. The transverse field mixes $|\uparrow\rangle$ and $|\downarrow\rangle$, so it changes total S^z. Equivalently,

$$[H_{\mathrm{TFI}}, S^z_{\mathrm{tot}}] \neq 0.$$

Therefore one should not impose `conserve="Sz"` for that model. In fact, TenPy's `SpinHalfSite` documentation says that with `conserve="Sz"`, the onsite operators `Sx`, `Sy`, `Sigmax`, and `Sigmay` are excluded. That is exactly because they carry nonzero S^z charge. A transverse-field Ising model needs such spin-flip operators, so it must use a site representation without U(1) S^z conservation, or at most a different symmetry such as a discrete parity if the model supports it.

Thus the rule is not

"purification should avoid `conserve="Sz"`."

The rule is

"use only symmetries that the Hamiltonian exactly preserves."

For the J1-J2 Heisenberg Hamiltonian in this project,

$$S_i^z S_j^z, \qquad S_i^+ S_j^-, \qquad S_i^- S_j^+, \qquad F_z S_i^z$$

all conserve total S^z. Hence `conserve="Sz"` is appropriate for that model.

12 Why doubled charge conservation does not remove physical sectors

The phrase "the purified MPS has total doubled charge zero" means

$$Q_{\mathrm{phys}} - Q_{\mathrm{aux}} = 0.$$

It does not mean

$$Q_{\mathrm{phys}} = 0.$$

It means

$$Q_{\mathrm{phys}} = Q_{\mathrm{aux}}.$$

The common value can be anything.

For example, a term with all physical spins up and all auxiliary spins up has

$$Q_{\mathrm{phys}} = +N, \qquad Q_{\mathrm{aux}} = +N,$$

so

$$Q_{\mathrm{phys}} - Q_{\mathrm{aux}} = 0.$$

This term is allowed. A term with all physical spins down and all auxiliary spins down has

$$Q_{\mathrm{phys}} = -N, \qquad Q_{\mathrm{aux}} = -N,$$

so it is also allowed. Likewise every intermediate magnetization sector is allowed.

Equivalently, the doubled-charge-zero space contains all operator blocks

$$\mathcal{H}^{(M)}_{\mathrm{phys}} \to \mathcal{H}^{(M)}_{\mathrm{phys}}$$

for every physical magnetization M. It excludes only off-diagonal operator blocks

$$\mathcal{H}_{\text{phys}}^{(M)} \to \mathcal{H}_{\text{phys}}^{(M')} \qquad (M \neq M').$$

For a Hamiltonian conserving S^z_{tot}, the thermal operator

$$e^{-\beta H}$$

is block diagonal in M. Hence the excluded off-diagonal blocks are exactly zero anyway. Nothing physical is lost.

The partition function is still

$$Z(\beta) = \sum_M \operatorname{Tr}_{\mathcal{H}_{\text{phys}}^{(M)}} e^{-\beta H_M}.$$

All physical magnetization sectors contribute to the trace.

13 Why physical observables are unaffected

Let

$$X_\beta = e^{-\beta H/2}.$$

The purification is the vectorized operator

$$|\Psi_\beta\rangle = |X_\beta\rangle\rangle.$$

For a physical observable O,

$$\frac{\langle \Psi_\beta | O | \Psi_\beta \rangle}{\langle \Psi_\beta | \Psi_\beta \rangle} = \frac{\operatorname{Tr}(X_\beta^\dagger O X_\beta)}{\operatorname{Tr}(X_\beta^\dagger X_\beta)} = \frac{\operatorname{Tr}(e^{-\beta H} O)}{\operatorname{Tr}(e^{-\beta H})}.$$

This is the ordinary thermal expectation value.

If H conserves S^z_{tot}, then X_β is block diagonal in physical magnetization:

$$X_\beta = \bigoplus_M e^{-\beta H_M/2}.$$

The doubled-charge-zero purified MPS stores all of these blocks at once:

$$|X_\beta\rangle\rangle = \bigoplus_M |e^{-\beta H_M/2}\rangle\rangle.$$

The thermal trace still sums over M. Thus the symmetry is not changing the ensemble; it is only removing operator blocks that symmetry says are exactly zero.

There is one useful consequence: if an observable changes total S^z, such as a single S_i^x, its thermal expectation value is zero in a U(1)-symmetric thermal state. TenPy also does not provide `Sx` or `Sigmax` as onsite operators when `conserve="Sz"`, because those operators carry nonzero charge. Charge-neutral observables such as H, S_i^z, $S_i^+ S_j^-$, and combinations forming transverse correlations can still be represented in symmetry-compatible ways.

14 A useful table

For N spins, consider a physical configuration with total physical charge

$$Q_{\text{phys}} = 2S^z_{\text{tot}} = M_Q.$$

In the infinite-temperature purification, it is paired with the same spin configuration on the auxiliary leg. Since the auxiliary leg has the opposite effective charge, the doubled charge is

$$Q_{\text{doubled}} = Q_{\text{phys}} + Q_{\text{aux,effective}} = M_Q - M_Q = 0.$$

physical configuration	Q_{phys}	$Q_{\text{aux,effective}}$	Q_{doubled}
all up	$+N$	$-N$	0
all down	$-N$	$+N$	0
half up, half down	0	0	0
any configuration	M_Q	$-M_Q$	0

All physical magnetization sectors are compatible with the same total doubled charge sector.

15 Why imaginary-time evolution preserves the conclusion

Your Hamiltonian is

$$H = \sum_{\langle i,j\rangle} J_1\left(S^z_i S^z_j + \frac{1}{2}S^+_i S^-_j + \frac{1}{2}S^-_i S^+_j\right) + \sum_{\langle\langle i,j\rangle\rangle} J_2\left(S^z_i S^z_j + \frac{1}{2}S^+_i S^-_j + \frac{1}{2}S^-_i S^+_j\right) + F_z \sum_i S^z_i.$$

Each term conserves total S^z:

$$[H, S^z_{\text{tot}}] = 0.$$

Therefore $e^{-\beta H/2}$ also conserves physical S^z. Starting from a purification that contains all physical S^z sectors, imaginary-time evolution evolves each sector independently:

$$|\Psi_\beta\rangle = \sum_M \left(e^{-\beta H_M/2} \otimes \mathbb{1}_{\text{aux}}\right) |\Psi^{(M)}_\infty\rangle.$$

No sector is thrown away. The final thermal trace is still

$$Z(\beta) = \sum_M \operatorname{Tr}_{\mathcal{H}^{(M)}_{\text{phys}}} e^{-\beta H_M}.$$

16 Mapping this to `finite_T_J1J2.py`

The relevant code structure is:

```
site = SpinHalfSite(conserve=conserve)
```

inside `models/model_J1J2.py`, and

```
psi = PurificationMPS.from_infiniteT(M.lat.mps_sites(), bc=bc)
```

inside `imag_apply_mpo`.

The first line says: use block-sparse tensors with a U(1) charge if `conserve="Sz"`.

The second line says: initialize the grand-canonical infinite-temperature purification.

The evolution operator is generated by

```
M.H_MPO.make_U(-d * dt, approx).
```

Since `make_U(t)` approximates e^{tH}, using negative real imaginary-time steps gives approximately

$$e^{-dtH}.$$

The purification state should evolve as

$$|\Psi_\beta\rangle = e^{-\beta H/2}|\Psi_\infty\rangle.$$

Thus if the state is evolved by dt in the ket, the physical density matrix corresponds to an inverse-temperature increase

$$\Delta\beta = 2dt.$$

That is why the code has

```
beta += 2.  * dt.
```

17 What would be a fixed-sector calculation?

A fixed-sector calculation would explicitly choose one physical total charge sector. Examples:

(a) Using TenPy's canonical purification constructor:

```
PurificationMPS.from_infiniteT_canonical(sites, charge_sector=...)
```

This is designed for a canonical infinite-temperature ensemble in a specified charge sector.

(b) Using exact diagonalization with a charge-sector projection:

```
ExactDiag(model, charge_sector=...)
```

(c) Initializing a pure product state such as all spins down. That would not be a thermal purification of the full Hilbert space.

Your current code does none of these.

18 When should `conserve="Sz"` not be used?

Use `conserve="Sz"` only if the Hamiltonian conserves total S^z. The current J1-J2 Heisenberg model with longitudinal field does conserve it.

Do *not* use `conserve="Sz"` if you add terms such as

$$h_x \sum_i S_i^x, \qquad h_y \sum_i S_i^y,$$

or any other term that changes total S^z. In that case,

$$[H, S^z_{\text{tot}}] \neq 0,$$

and the U(1) symmetry assumption is invalid. Then use

```
conserve=None.
```

19 Why `conserve=None` and `conserve="Sz"` can differ numerically

In exact arithmetic and with sufficiently large bond dimension, the two calculations should give the same physical grand-canonical thermal observables for this Hamiltonian:

`conserve=None` and `conserve="Sz"`.

However, with finite bond dimension and truncation, they can differ slightly. Reasons include:

- The block-sparse calculation distributes the available bond dimension across symmetry sectors.
- The non-symmetric calculation has no sector structure, so truncation is organized differently.
- At the same nominal χ_{max}, the effective variational spaces are not identical.
- Numerical roundoff and singular-value truncation can differ.

Such differences are truncation effects, not evidence that `conserve="Sz"` computes a fixed physical magnetization ensemble.

20 Practical checklist for the current code

For the present J1-J2 triangular Heisenberg model:

- `conserve="Sz"` makes sense.
- The ensemble initialized by `from_infiniteT` is grand-canonical in physical magnetization.
- The physical trace includes all S^z_{tot} sectors.
- `conserve=None` is allowed but should be slower and usually less efficient.
- `conserve=None` is required only if the Hamiltonian breaks total S^z conservation.

Two unrelated code-cleanup points:

- The loop `while beta < beta_max` can overshoot by one step due to floating-point roundoff. An integer step count is safer.
- If the plotted C_v is meant to be specific heat per site, divide the finite-difference result by $N = L_x L_y$. Without that division, it is the total heat capacity of the finite cluster.

Final statement

The phrase `conserve="Sz"` is easy to misread. In this purification context it should be read as:

"use S^z conservation as a tensor symmetry."

It should not be read as:

"fix the physical thermal ensemble to one S^z sector."

For the current J1-J2 Heisenberg Hamiltonian, using `conserve="Sz"` in the purification calculation is physically correct.